Memory-Defined Phase Maps for Dissipative Quantum Reservoirs

Anonymous QCE Submission

***Abstract*—We study when a stateful dissipative gate-model quantum reservoir is useful for scalar time-series prediction. The paper's central object is a phase map, not a single tuned benchmark point. Across four tasks and ten reservoir seeds, good validation performance concentrates in a reproducible operating regime with gentle input drive, nonmaximal recurrent coupling, and controlled damping. We formulate this as a universal operating-regime hypothesis for this QRC family: learnable temporal tasks are expected to occupy the same memory-preserving band, rather than arbitrary coordinates of the reservoir parameter space. Leave-one-task and leave-one-seed tests, mechanism ablations, and intrinsic diagnostics support the hypothesis. The result is empirical and simulator-level; it is not a theorem over all possible data-generating processes or a hardware-resource claim.**

***Index Terms*—quantum reservoir computing, dissipative dynamics, memory capacity, phase diagrams, operating regimes**

I. Question and Claim

Reservoir computing trains a linear readout on fixed nonlinear dynamics [1], [2]. In a quantum reservoir, Hilbert-space dimension alone is not the useful resource. The dynamics must preserve recent input history, forget old history, and expose nonlinear observables to the readout [4], [5], [6]. Too much input drive overwrites state, zero damping prevents controlled fading memory, and missing recurrent mixing leaves weak temporal features.

We therefore treat the phase diagram as the scientific object. The claim is a QRC-only generalization claim: for the studied stateful dissipative QRC family, useful learning is organized by a transferable operating regime. The strongest reviewer-safe statement is a hypothesis supported by four tasks, ten seeds, and ablations: scalar temporal tasks that can be learned by this architecture should be searched first in a low-input, nonzero-damping, nonmaximal-coupling band.

II. Protocol

The reservoir carries a density matrix from one input time step to the next. For scalar input u_t,

$$\rho_{t,0} = U_{\mathrm{in}}(u_t)\rho_{t-1,L}U_{\mathrm{in}}^{\dagger}(u_t), \qquad U_{\mathrm{in}}(u_t) = \bigotimes_i R_y(\beta u_t). \tag{1}$$

Each virtual layer applies frozen local rotations, a ZZ ring, and a product noise channel:

$$\rho_{t,\ell} = \mathcal{D}_{\gamma}^{\otimes n}\left[U_\lambda U_{\mathrm{mix}}\rho_{t,\ell-1}U_{\mathrm{mix}}^{\dagger}U_\lambda^{\dagger}\right], \qquad U_\lambda = \prod_{(i,j)\in E_{\mathrm{ring}}} e^{-i\lambda Z_i Z_j}. \tag{2}$$

The main phase maps use $n = 4$, two layers, ring topology, uniform input, amplitude damping, the $Z+ZZ$ QRC16 readout, and ridge-only training. Hyperparameters are never selected by holdout data.

For task j and seed s, let $r_{j,s}(\theta)$ be the validation percentile rank of $\theta = (\beta, \lambda, \gamma)$, with lower rank better. A transferable operating band is

$$B_{p,q} = \{\theta : \Pr_{j,s}[r_{j,s}(\theta) \le p] \ge q\}. \tag{3}$$

This frequency-level definition is deliberately weaker than a universal optimum. It asks whether a region repeatedly behaves well, not whether the same coordinate wins every task.

III. Evidence

Phase-map transfer. Fig. 1 shows the core pattern: low validation rank is concentrated, not randomly scattered. The primary band $B_{20,0.7}$ has 4 connected points, all at $\gamma = 0.12$. When one task is withheld while defining the band from the remaining tasks, the held-out task-seed mean validation rank is 0.240 with bootstrap CI $[0.215, 0.265]$. When one seed is withheld, the corresponding mean rank is 0.180 with CI $[0.154, 0.209]$. The band is therefore a top-quartile search prior that transfers across both task identity and reservoir initialization.

Mechanism stress tests. The operating band is not just a plotting artifact. At the central slice, the projected base band has 4 points. Removing the controlled damping mechanism gives retention 0.00; removing recurrent mixing gives retention 0.00; replacing amplitude damping by dephasing gives retention 0.00. In every focused ablation, the base band ceases to be a robust top-ranked region under the same validation-rank criterion. The result supports the mechanism story: useful dynamics need input that is not too strong, recurrent mixing, and damping that preserves usable memory rather than merely adding noise.

Variant	Band pts.	Retention	Rank on base band
base AD+RxZZ	4	1.00	0.256
$\gamma = 0$	0	0.00	0.465
dephasing	0	0.00	0.442
depolarizing	0	0.00	0.526
no mixer	0	0.00	0.403
Rx only	0	0.00	0.521
ZZ only	0	0.00	0.503

Diagnostics. Memory-based diagnostics identify the same useful region before task-specific holdout evaluation. On the current simulator-backed diagnostic rerun, MC has Spearman $\rho_s = -0.85$ against validation rank; $\mathrm{IPC}_{\mathrm{mem}}$, $\mathrm{IPC}_{\mathrm{tot}}$, and $\mathrm{IPC}_{\mathrm{nonlin}}$ give -0.84, -0.90, and -0.80. Feature variation and

TABLE I
QRC-ONLY AUDIT TRAIL. HOLDOUT DATA ARE NOT USED TO DEFINE THE OPERATING REGIME.

Item	Value
Tasks and splits	Mackey–Glass, Lorenz, NARMA10: washout/train/validation/holdout $= 300/1600/400/400$; annual Sunspots: $20/150/50/60$. Inputs are scaled to $[-1, 1]$ and targets standardized.
Seeds and ridge	Reservoir seeds 42–51; ridge grid $\{10^{-8}, 10^{-7}, \ldots, 10^{3}\}$ selected on validation.
Phase grid	QRC16 coarse grid over $\beta/\pi = \{0, 0.02, 0.04, 0.07, 0.10, 0.22, 0.50\}$, $\lambda/\pi = \{0, 0.05, 0.10, 0.12, 0.16, 0.28, 0.42\}$, $\gamma = \{0, 0.05, 0.12, 0.22, 0.30\}$.
Primary band	$B_{20,0.7}$ contains 4 connected points; the medoid is $(\beta/\pi, \lambda/\pi, \gamma) = (0.04, 0.12, 0.12)$. Relaxing to $B_{20,0.6}$ gives 14 points and $B_{30,0.7}$ gives 46 points.
Stress tests	Focused QRC-only ablation maps at $\gamma = 0.12$ compare base amplitude damping against $\gamma = 0$, dephasing, depolarizing, no mixer, Rx-only mixer, and ZZ-only mixer. The readout, tasks, seeds, and ridge protocol are unchanged.
Diagnostics	Memory capacity and intrinsic processing capacities are computed from simulator trajectories and compared to validation rank and top-decile screening retention.

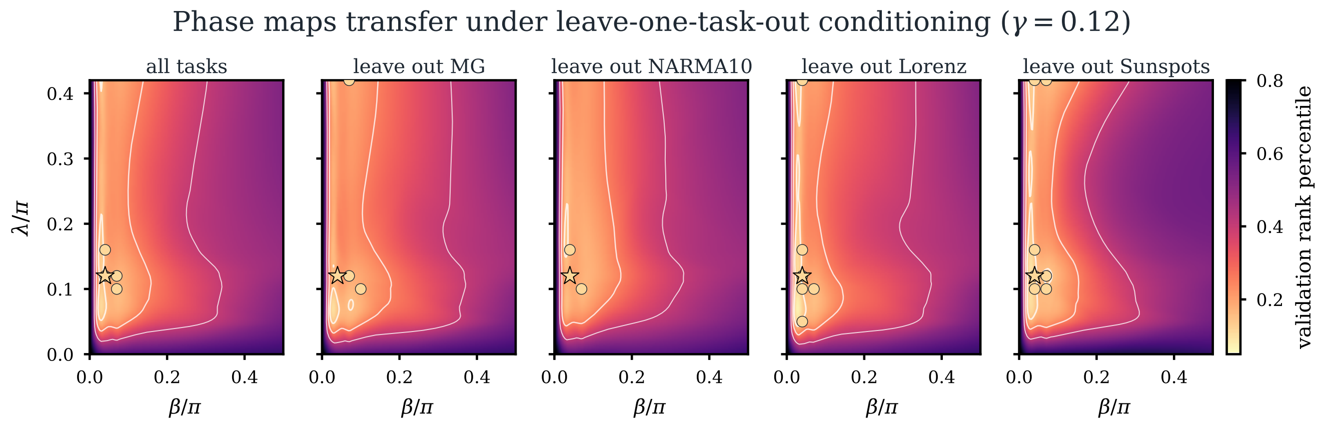

Fig. 1. Validation phase maps at the fixed $\gamma = 0.12$ slice. Bright regions are low validation percentile rank. Gold circles mark points that appear in the top-20% for at least 70% of the displayed task-seed replicates; the star is the medoid of the primary band. The same low-input, nonmaximal-coupling region persists under leave-one-task-out conditioning.

effective rank are weakly positive, 0.16 and 0.15, so raw diversity alone is not the mechanism.

IV. Discussion and Scope

The evidence supports a universal operating-regime hypothesis, not a theorem. The tested tasks differ in chaotic smooth dynamics, nonlinear memory, and real annual observations, and the same frequency-defined phase region remains useful across seeds and leave-one-task-out splits. The statement should still be read within the modeled family: four-qubit stateful gate-model QRCs, scalar time-series prediction, exact simulated expectation values, ridge readouts, and the specified grid.

The scope is intentionally narrower than a hardware or advantage claim. Finite-shot sampling, measurement-basis scheduling, calibration drift, and device noise are not modeled. Likewise, the phase map does not say that every task has the same optimum. It says that useful QRC learning should first be searched in a memory-preserving band, and that leaving this band by removing damping or recurrent mixing destroys the robust phase-map signature.

V. Conclusion

Dissipative QRC performance is organized by a memory-defined operating regime. The phase maps transfer across tasks and seeds; mechanism ablations remove the robust base band; and intrinsic memory/processing diagnostics recover the same useful region. For this QRC family, the practical claim is simple: before scaling features or chasing a benchmark winner, identify the phase-map band where input drive, recurrence, and damping jointly preserve usable temporal memory.

References

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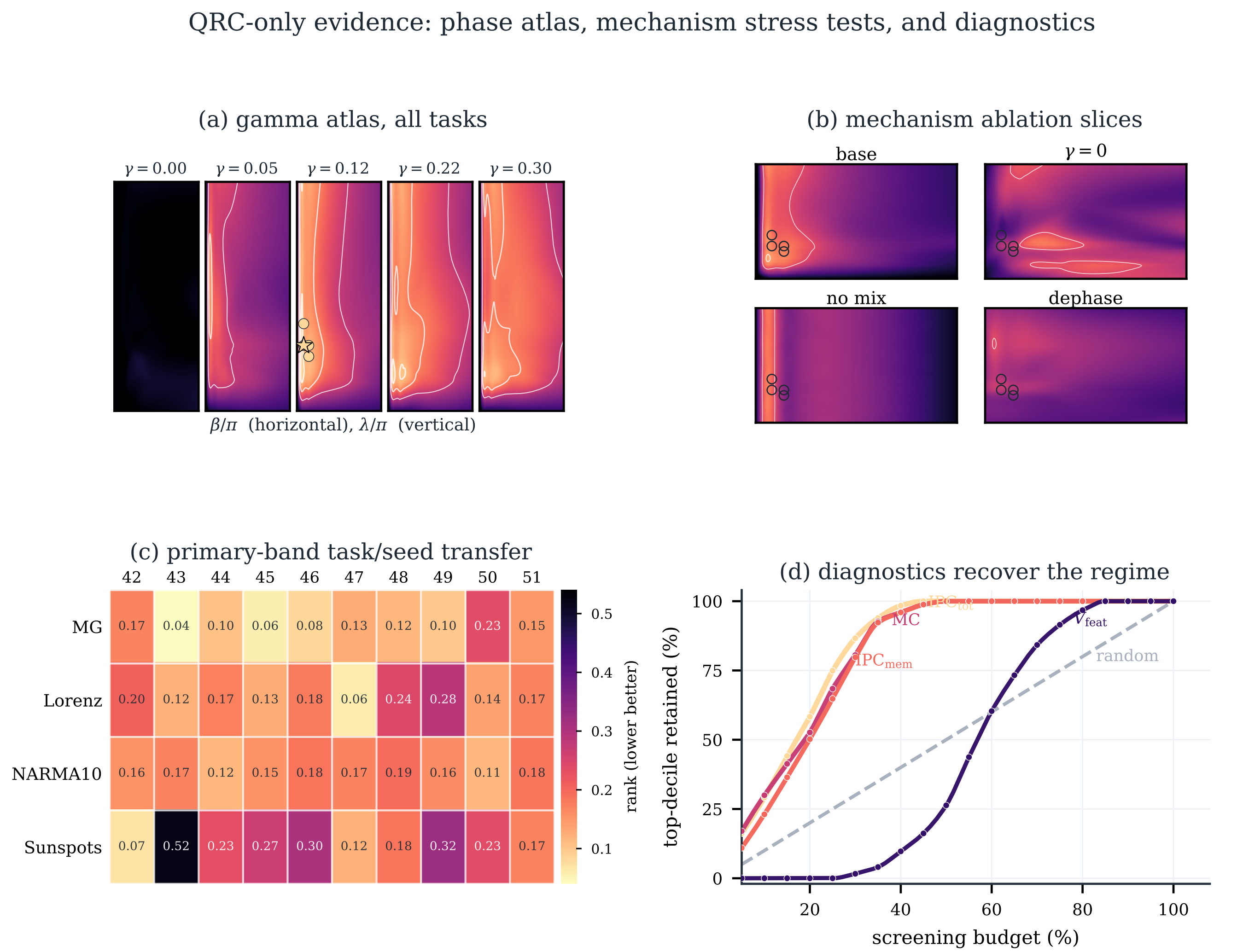

Fig. 2. QRC-only evidence for the operating-regime hypothesis. (a) Gamma-slice atlas: the useful region emerges only away from zero damping and away from maximal coupling. (b) Focused ablations at $\gamma = 0.12$: outlines show the base $B_{20,0.7}$ projection; $\gamma = 0$, no-mix, and dephasing controls do not preserve it as a robust top region. (c) Task-seed transfer ranks for the primary band. (d) Intrinsic memory and processing-capacity diagnostics recover top validation-decile reservoirs more efficiently than random or diversity-only screening.

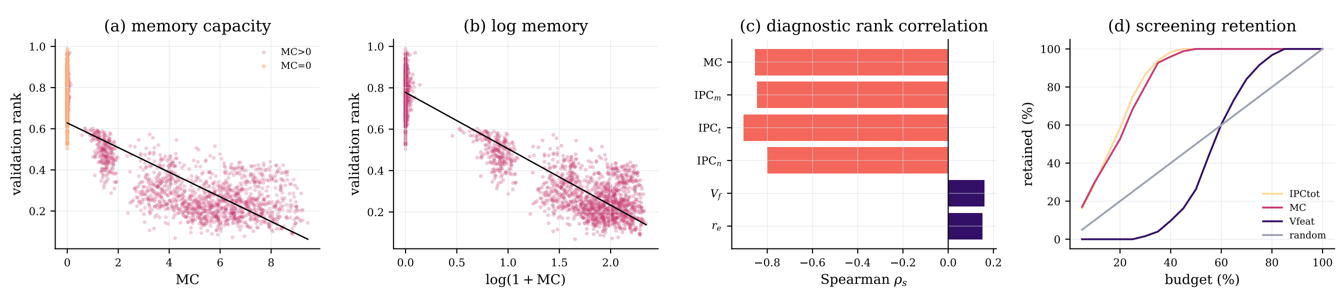

Fig. 3. Diagnostic audit for the QRC phase-map claim. Raw and log-scaled memory capacity both anticorrelate with validation rank, intrinsic processing-capacity diagnostics give the strongest rank correlations, and screening by memory/IPC recovers useful reservoirs earlier than random or diversity-only screens.

TABLE II
ARTIFACT MAP FOR EXACT REPRODUCTION OF THE QRC-ONLY PAPER.

Artifact	Role
QRC phase grid	`data/qrc_seed_ensemble_grid.csv`: main QRC16 phase-map grid over tasks, seeds, and (β, λ, γ).
QRC ablation slices	`data/qrc_phase_ablation_slice_grid.csv`: focused mechanism stress tests at $\gamma = 0.12$.
Band statistics	`data/phase_map_generalization_stats.json`: band sizes, connectedness, leave-one-task/seed bootstrap CIs, ablation retention, and diagnostic correlations.
Generated numbers	`paper/generated/phase_map_numbers.tex`: all numeric macros used in the manuscript.
Figures	`scripts/make_figures_and_build_data.py`: regenerates the phase maps, ablation atlas, transfer matrix, and diagnostic screening plots.
Build	`./reproduce.sh` and `./paper/build.sh --update-pdf`: end-to-end regeneration and PDF refresh.